

Models from experimental data

R0005E - Measurement and Feedback Control

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Model from experimental data

If one can create a model from first principle, why bother to directly find a model from experiments?

Remember: We want to control the process, so that a desired behavior is attained!

Some reasons:

- If not all parameters are known or can be directly assessed, then experiments could help to find the missing parameters.
- The process is too complex and there is little knowledge or understanding to derive a model.
- Any model usually contains many approximation. Thus, it needs to be validated through experiments and comparisons. As a result, the model and its parameters can be refined.
- Any system is subject to changes over its life-cycle. A model need to adapt to these changes. Such changes are rarely assessable from 1st principle.

The model generation task $\triangleq$ system identification:

1. Conduct an experiment with the system that reveals its behavior.
2. Acquire data for inputs and outputs at the same time.
3. Determine a model structure, e.g. type of transfer function.
4. Derive the parameters of the model from one data set.
5. Validate the model using another data set.

System identification is an advanced topic. At LTU, it can be given as a graduate course.

Depending on the type of experimental data that is used, different methods are available to derive models:

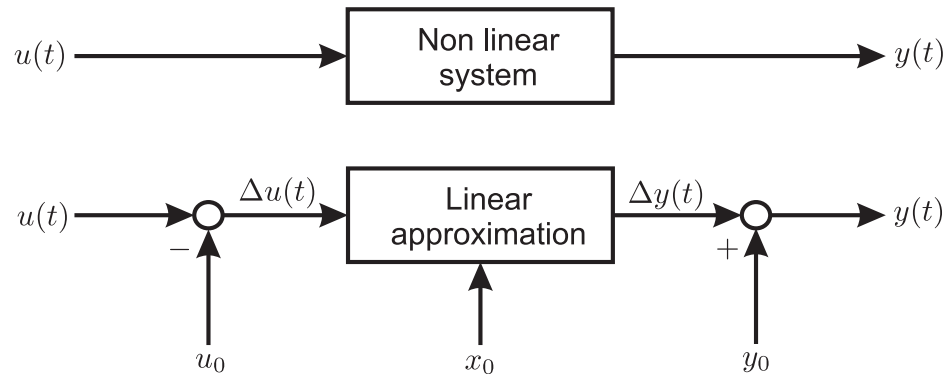
- Transient response data:
We assume that the system is well described by a transfer function and we conduct a step or impulse response experiment on the system.
- Frequency response data:
We choose a set of frequencies for a sinusoidal input and observe how well the system can follow that signal. From this information a model can be composed.
- Stochastic steady-state information:
We observe the process in operation and derive the DC-Gains between inputs and outputs. This is usually done in multivariate analysis and modeling.
- Pseudorandom noise information:
Using mathematical methods a model is derived that can reproduce the output noise sequence from the input noise sequence. The result can be a transfer function.

Here we will only have a look into **transient response data**.

Experimental data

We will only have a look at **step response data**. This is probably the most commonly used experimental data type.

Important note: Since we want to derive a transfer function from a step response, we have to remember that the model that we derive is a linear model, which is only valid around an operating point!



Consequence: We have to remove the operating point from the input and output data, so that the data represents the step response in terms of $\Delta u(t)$ and $\Delta y(t)$.

This step is very important if a computer tool is used for the system identification, but it is usually not a problem when it is done by hand.

Conducting the experiment and acquiring data are very crucial steps in the modeling. In industry, they might be very expensive and re-doing them is not always an option. Planning is therefore very important.

The quality of the acquired data limits the achievable quality of the generated model. It is therefore important to have a good experimental design.

Some guidelines:

- Run the experiment in open-loop.
This means that your input signal to the system is not affected by other signals that correlate with the output signal that you acquire.
- Better to sample fast.
Data that you acquire in the experiment is sampled by a computer. If you sample at a relatively slow rate, then there will be negative effects on the data quality. You might also not be able to observe the transient correctly. This can not be corrected afterwards.
- Disturbances and other exogenous inputs.
Try to gather as much information as possible on other inputs to the system and how they affect your experiment.
- Operational data.
In some case, data that is acquired during normal operation and can yield good models. But be careful, model validation is crucial and understanding all aspects of normal operation is important. Operational data usually gives a good hint on the model structure.

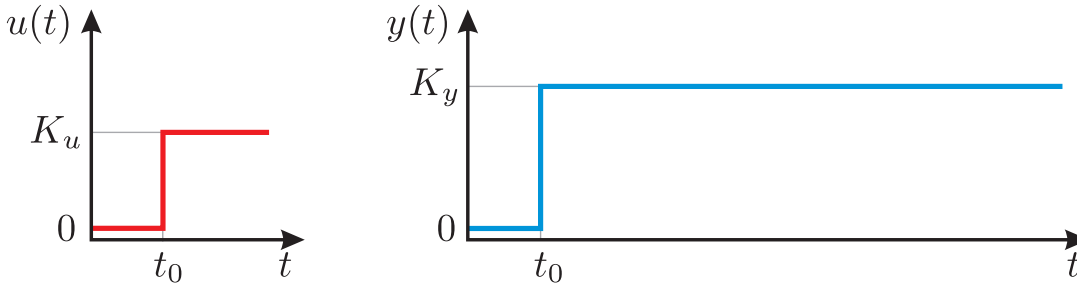
Important notes:

- Downsampling.
Data that is acquired at a high sampling rate can be sampled down to a slower rate.
- Noise.
Production hardware might have more noise than the experimental hardware. You can add noise to your data to simulate production hardware properties.
- Quantization.
Production hardware might have worse resolution than the experimental setup. This can be simulated as well.

Model structure and parameter calculation

After data from a step response experiment are available, the data can be examined and the model structure need to be determined.

Step response data: Static gain



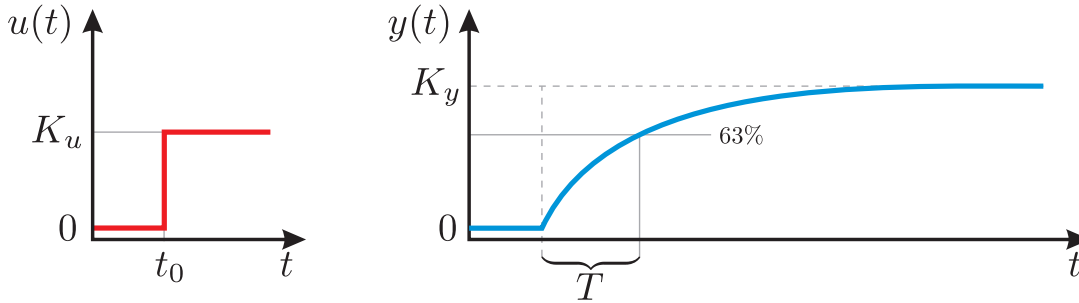
Assumed model structure:

$$H(s) = K$$

Parameter calculation:

$$K = \frac{K_y}{K_u}$$

Step response data:



Assumed model structure: First order system.

$$H(s) = \frac{K}{Ts + 1}$$

Parameter calculation:

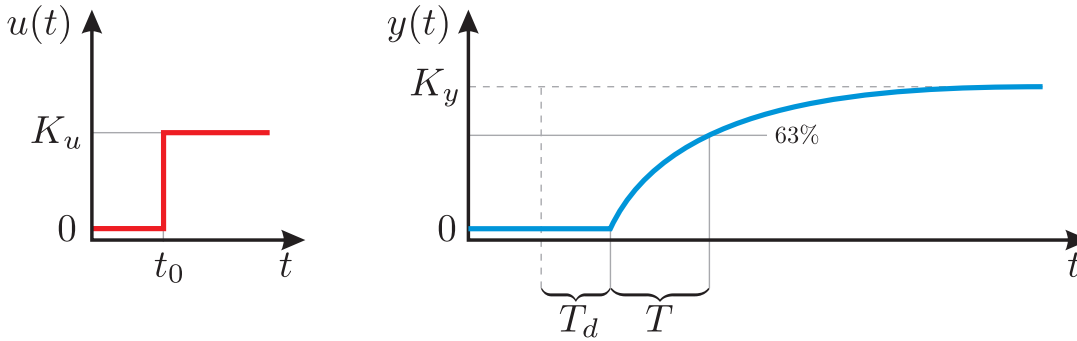
- DC-Gain:

$$K = \frac{K_y}{K_u}$$

- Time constant:

The time constant is found by calculating the time between t_0 and until the response has reached $y(t) = 0.63K_y$.

Step response data:



Assumed model structure: First order system with time delay

$$H(s) = \frac{K e^{-T_d s}}{T s + 1}$$

Parameter calculation:

- DC-Gain K :

$$K = \frac{K_y}{K_u}$$

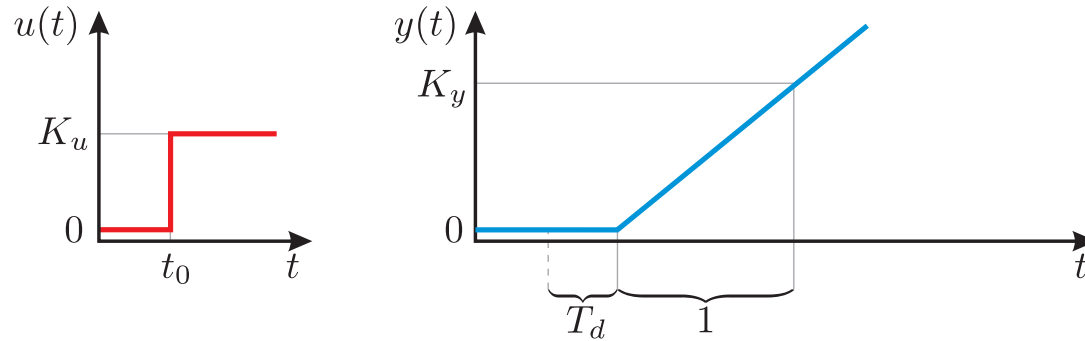
- Time delay T_d :

Determine the time between t_0 and until $y(t)$ starts to rise from 0.

- Time constant T :

The time constant is found by calculating the time between $t_0 + T_d$ and until the response has reached $y(t) = 0.63K_y$.

Step response data:



Assumed model structure: Integrator with time delay

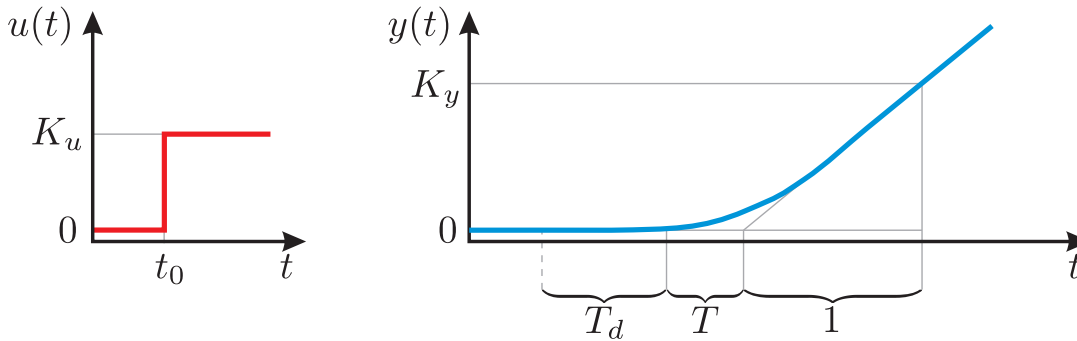
$$H(s) = \frac{K e^{-T_d s}}{s}$$

Parameter calculation:

- Time delay T_d :
Determine the time between t_0 and until $y(t)$ starts to rise from 0.
- Gain constant K :
The gain constant is the slope of the response curve $y(t)$. It could be determined 1sec after $y(t)$ rises from 0 as

$$K = \frac{K_y}{K_u}$$

Step response data:



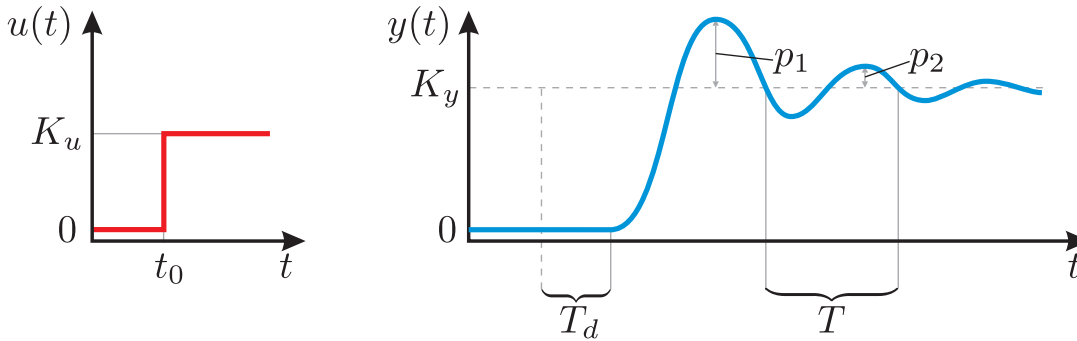
Assumed model structure: First order system with Integrator and time delay

$$H(s) = \frac{K e^{-T_d s}}{s(Ts + 1)}$$

Parameter calculation:

- Time delay T_d :
Determine the time between t_0 and until $y(t)$ starts to rise from 0.
- Time constant T :
The time constant is the time between $t_0 + T_d$ and where the slope line intersects with the 0 level.
- Gain constant K :
The gain constant is the slope of the response curve $y(t)$ as before.

Step response data:



Assumed model structure: Oscillating second order system with time delay

$$H(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} e^{-T_d s}$$

Parameter calculation:

Here we use the logarithmic decrement to calculate ζ and ω_n . This is also discussed in Problem 3.32.

- Time delay T_d :
Determine the time between t_0 and until $y(t)$ starts to rise from 0.
- DC-Gain K :
The DC-Gain is given as the ratio $\frac{K_y}{K_u}$.

- 1. Calculate the ratio $d = \frac{p_2}{p_1}$
- 2. Determine ζ from

$$\zeta = \frac{1}{\sqrt{\left(\frac{2\pi}{\ln d}\right)^2 + 1}}$$

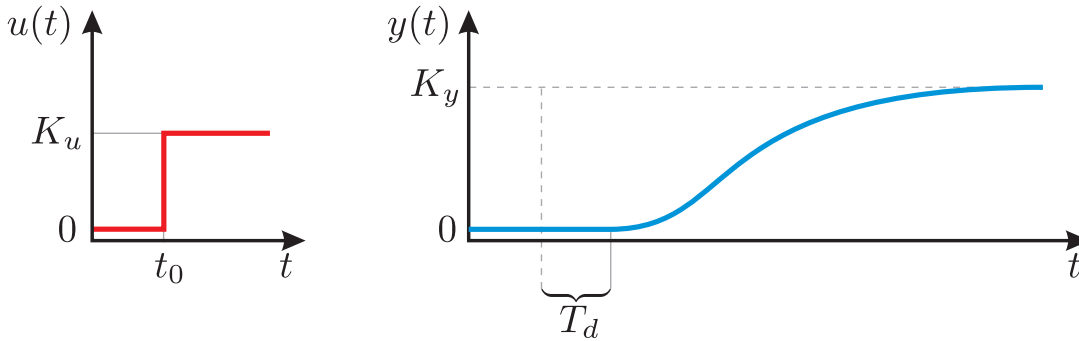
- 3. Derive ω_n from

$$\frac{2\pi}{T\sqrt{1 - \zeta^2}}$$

Alternatively, p_1 is the overshoot and ζ can be calculated from it. Using the damped frequency ω_d , the natural frequency ω_n can be calculated.

Note: It is very hard to distinguish when $y(t)$ rises from 0, in order to determine the time delay. This creates uncertainties in the calculated parameters!

Step response data:



Assumed model structure: Second order system with real poles and time delay (potentially a zero as well)

$$H(s) = \frac{K(s + b_0)}{(s^2 + a_1s + a_0)}e^{-T_d s}$$

Parameter calculation:

In this case, only an iterative procedure can be used. There is no straight forward one-shot solution.

First calculate the time delay T_d and the gain constant K :

- Time delay T_d :
Determine the time between t_0 and until $y(t)$ starts to rise from 0.
- DC-Gain K :
The DC-Gain is given as the ratio $\frac{K_y}{K_u}$.

The remaining transfer function to be generated is

$$H(s) = \frac{s + b_0}{(s^2 + a_1s + a_0)}$$

This also implies that we need to adjust the data set $y(t)$ to become $y'(t) = \frac{y(t)}{K}$.

Thereafter, assume that the time series is a superposition of elementary time series, like in the partial fraction expansion:

$$y'(t) = y'(\infty) + Ae^{-\alpha t} + Be^{-\beta t}$$

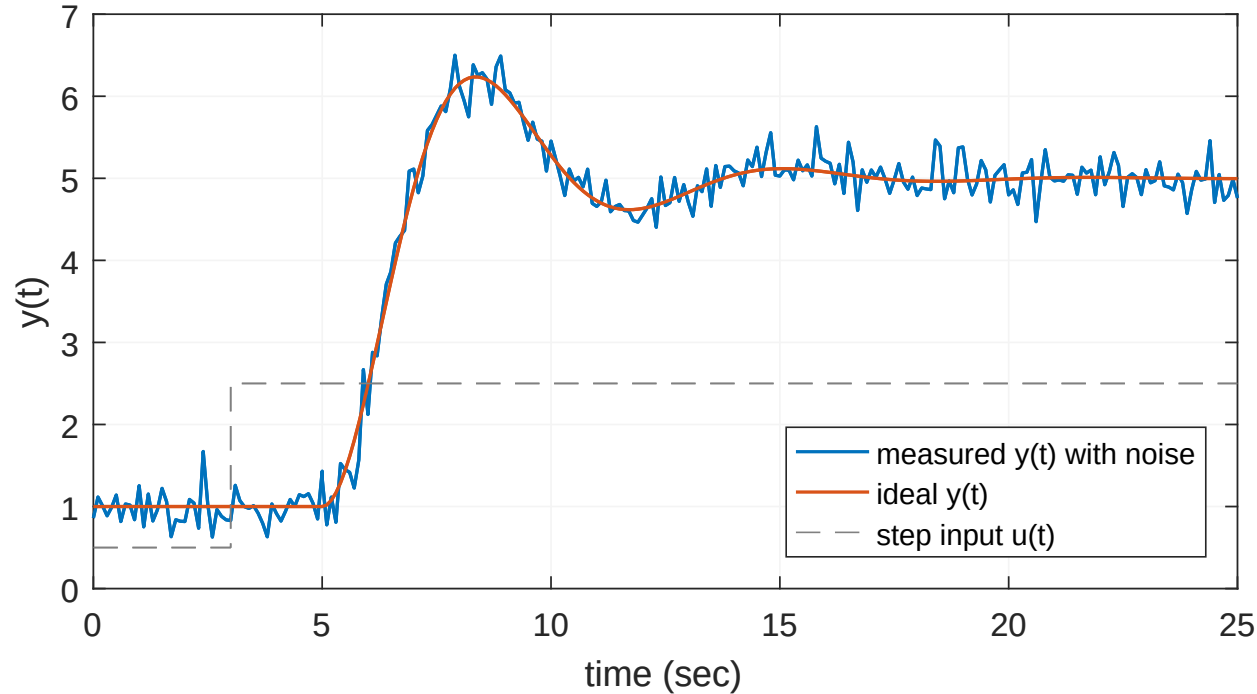
1. Assume $y'(t) - y'(\infty) \approx Ae^{-\alpha t}$.
2. Calculate $r(t) = \log_{10}(y'(t) - y'(\infty))$.
3. Perform a linear fit to the data $r(t) = \mu t + \nu$. μ needs to be negative.
4. Derive A and α from
$$\log_{10} A = \nu, \quad \alpha t \log_{10} e = \mu.$$
5. Assume $y'(t) - y'(\infty) - Ae^{-\alpha t} \approx Be^{-\beta t}$
6. Redo the steps from above.
7. Calculate the transfer function from the Laplace transform of the elementary terms.

Note: This procedure is very sensitive to numerical uncertainties. Be aware that the second term might be calculated from rather small number. Noise will have a negative effect on the result.

Model quality

In any experiment there will be noise and disturbances acting on the system that you want to understand.

Look at the following noisy step response:



This is the response of the following system:

$$H(s) = \frac{e^{-2s}}{s^2 + 0.7s + 1}$$

Observations:

- The oscillation is hard to be seen. At least the second peak is not visible.
- The peak is uncertain.
- The time delay is rather hard to identify exactly.

Important note: Choose always a step size for the input signal so that there is a large signal to noise ratio!

Additional material

System identification is an advanced topic and in this course we only touch the very basic aspects and take a very practical approach. Nevertheless, the following video is interesting in that context:

- System Identification Methods

<https://www.youtube.com/watch?v=YiJRgKJ0My8>

In that video there are many references to the frequency domain which will be discussed in later lectures. So, you might not understand all the details, but overall, the video presents the basic notions and provides a different view point.